

POS 6933 | Spring 2026

Modeling Guide

How to build, run, and evaluate a theory-driven forecasting model

This guide walks you through the process of moving from a theoretical idea to a complete forecasting workflow. For illustration, it uses democracy as the example theory. It also includes a sample memo, `03_Democracy_Model_Memo`, to show one way you might organize and present your results. The example here is intended purely for demonstration and should not be taken as a model of strong work.

Prepared for the Conflict Forecasting Project

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1. Purpose

This guide explains how to build a theory-driven forecasting model, estimate it with two machine-learning algorithms, compare it to the two benchmark models, and write a memo based on the results. The Democracy example is a worked illustration of the standard expected in the project.

2. What You Are Building

For the final modeling assignment, every student or team must do four things:

1. choose one theoretical model from the *Model Development* handout,
2. operationalize that theory with variables already available in `data/imputed\_dataset.rds`,
3. estimate the theory with **two algorithms**, and
4. compare the theory-driven models to the benchmark models using out-of-sample metrics.

The final deliverable is a **7–10 page analytical memo**. The memo should read like a short research paper, not like a loose collection of output files.

A strong project shows a clear chain of reasoning:

- why the theory should matter for conflict forecasting,
- why the selected variables represent that theory,
- why the chosen algorithms are appropriate,
- whether the theory improves prediction relative to the benchmarks.

3. How the Project Folder Works

This student project folder has been organized so that each folder has one clear purpose and make the process of model building straightforward.

Folder or file	Purpose
01_Model_Development.pdf	Assignment brief explaining the theoretical-model task, algorithm choices, evaluation criteria, and memo expectations.
02_Modeling_Guide.pdf	This guide. It walks students through a worked example.
data/	Main modeling dataset. Students should begin here when checking which variables are available.
codebooks/	Documentation for variable definitions, coding rules, and concepts.
benchmark_results/	Benchmark prediction files, benchmark tuning tables, and benchmark comparison plots from earlier in the course.
code/	R script that estimates the Democracy example and writes output files.
results/	Generated output folder for the models, comparison tables, and plots.

4. Recommended Workflow

You may benefit from this recommended workflow for your project.

4.1. Step 1: Choose a theory

Start with 01_Model_Development.pdf. Pick **one** theoretical model from that handout.

4.2. Step 2: Map the theory to variables

Explore the imputed dataset and review the variable names carefully. Then consult the relevant codebooks to identify indicators that substantively fit your chosen theory. These measures do not need to map onto the theory perfectly; in many cases, theoretically defensible proxies are entirely appropriate.

The script below shows how you need to read the dataset into R.

```
demo_data <- readRDS("data/imputed_dataset.rds")

dim(demo_data)
grep("^(vdem_|wgi_|polity_|fh_)", names(demo_data), value = TRUE)[1:40]
```

4.3. Step 3: Estimate the theory with two algorithms

In this demo project, the example script fits:

- a Random Forest model, and

- a Gradient Boosting Machine model.

4.4. Step 4: Compare to the benchmarks

Your theory-driven models are not evaluated in isolation. They must be compared to the benchmark models already saved in `benchmark\_results/`. The core evaluation metrics are:

- Mean Squared Error (MSE),
- Root Mean Squared Error (RMSE),
- Continuous Ranked Probability Score (CRPS),
- calibration and sharpness of predictive intervals.

5. An Example: Democracy Model

5.1. The idea

For this worked example, I want to know if democracy or lack of it as a theory shape the likelihood and intensity of conflict.

5.2. Operationalization strategy

The Democracy model below is a **theory-based specification**. It is not just a random collection of regime variables. Each indicator might capture one dimension of democratic practice or institutional constraint. The table below presents the variables chosen.

Variable	Source	Why it belongs in the Democracy specification
<code>vdem_v2x_polyarchy</code>	V-Dem	Captures electoral contestation, inclusion, and the procedural core of democracy.
<code>vdem_v2x_libdem</code>	V-Dem	Captures liberal constraints, checks on power, and rights protection.
<code>vdem_v2x_partipdem</code>	V-Dem	Captures participatory inclusion and broader channels for political voice.
<code>vdem_v2x_delibdem</code>	V-Dem	Captures the quality of public reasoning and collective decision-making.
<code>vdem_v2x_egaldem</code>	V-Dem	Captures whether access to political influence is distributed more equally.
<code>wgi_VA_EST</code>	WGI	Measures voice and accountability.
<code>wgi_GE_EST</code>	WGI	Measures government effectiveness, which conditions how democratic institutions function in practice.
<code>wgi_RL_EST</code>	WGI	Measures rule of law and the stability of legal constraints.
<code>fh_c1</code>	Freedom House	Measures civil liberties and nonviolent channels for organization and expression.
<code>polity_polity2</code>	Polity	Provides a broad regime anchor that captures overall democratic structure.

5.3. How to justify the variables in your memo

In the final memo, do not justify variables by saying only that they were available in the dataset. Instead, do two things:

1. explain what each variable measures, and
2. connect that measurement to a substantive argument in the literature.

The table above shows the logic of the Democracy model, but your final memo still needs real scholarly citations to support your chosen variables.

6. How To Run The Example Code

6.1. The script

The full R code pipeline is in:

```
code/model\_building\_tutorial.R
```

6.2. What the script does

The script is designed to do the full example workflow from start to finish. It:

1. reads the imputed dataset,
2. builds horizon-specific outcomes,
3. fits the Democracy theory with Random Forest and GBM,
4. loads the benchmark prediction and tuning files,
5. matches predictions on the same rows,
6. computes comparison metrics,
7. writes tables, model summaries, and plots to the `results/` folder.

6.3. Where the outputs go

After the script finishes, look in:

```
results/democracy\_model/
```

That folder should contain four main pieces:

- `model\_outputs/` for the two Democracy models,
- `comparison/` for benchmark comparison tables,
- `plots/` for polished figures,

7. What A Strong Memo Looks Like

Your final memo should be about 7–10 pages and should read as a short publishable research paper. A strong memo usually includes the following sections:

1. **Introduction and theory choice** Explain the theory, why it matters for conflict forecasting, and what you expect it to do.
2. **Operationalization** List the indicators, explain what they measure, and justify why they belong in the specification.
3. **Data and design** Briefly explain the outcome, forecast horizons, train/test design, and benchmark comparison strategy.
4. **Algorithms** Explain the two algorithms you estimated and why they are reasonable choices for this theory.

5. **Results** Use clear tables and publication-quality plots to compare your models with the benchmarks.
6. **Interpretation** Explain what the results imply about your theory, not just which model had the lowest error.
7. **Conclusion** State clearly whether the theory appears useful for forecasting and what its limitations are.

8. Common Mistakes To Avoid

- Choosing variables because they are convenient rather than theoretically appropriate.
- Listing variables without explaining what each one captures.
- Describing model output without comparing it to the benchmarks.
- Reporting only one metric and ignoring the others.
- Writing a memo that reads like a code log instead of an research paper.

9. Final Checklist

Before you submit, confirm that you can answer “yes” to each question below.

- Did I choose one theory and explain why it should matter for conflict forecasting?
- Did I justify each indicator with substantive logic and literature?
- Did I estimate two algorithms?
- Did I compare my models to both benchmark models?
- Did I use MSE, RMSE, and CRPS in the evaluation?
- Did I include polished plots and clear tables?
- Did I write a memo that explains what the results mean?